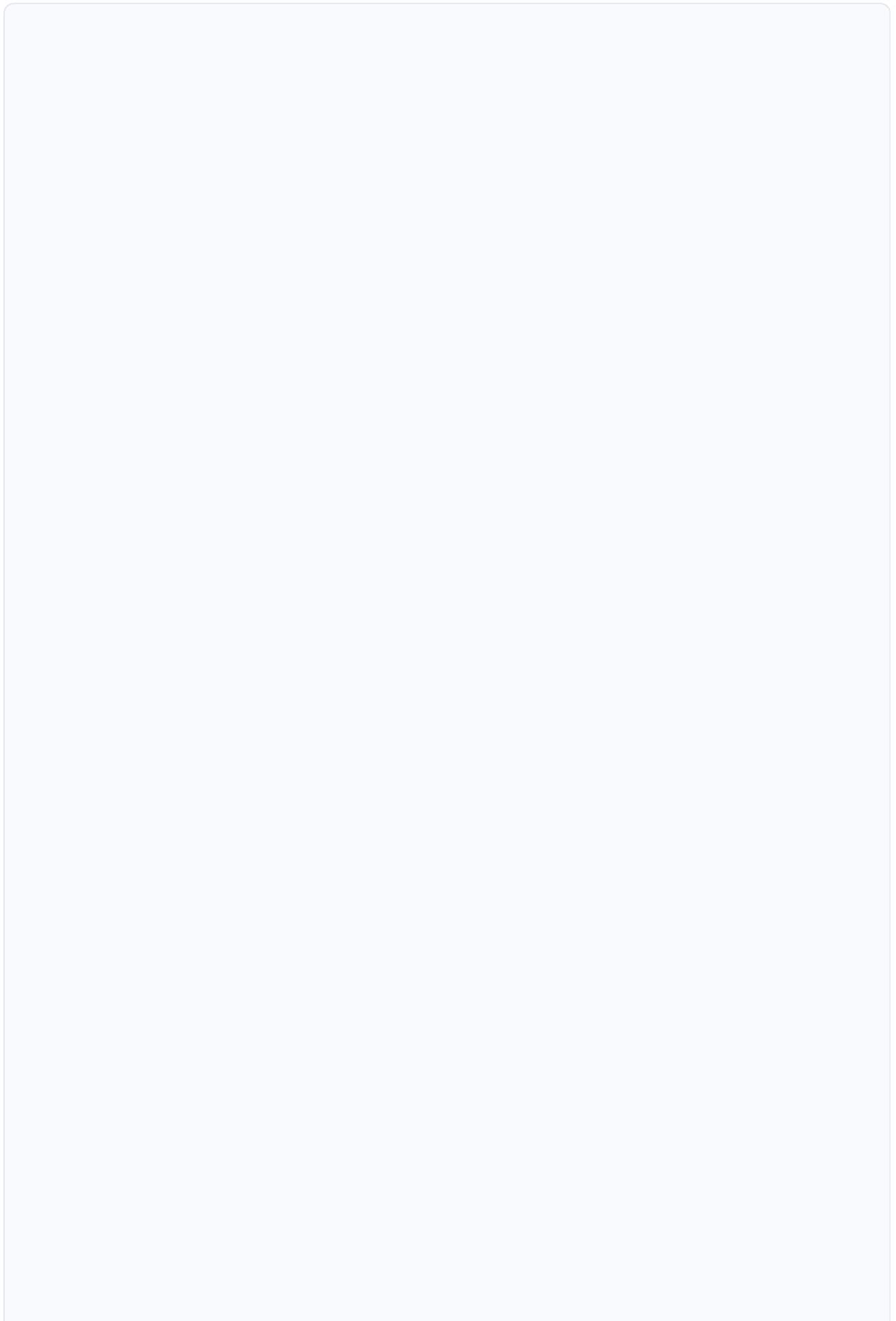


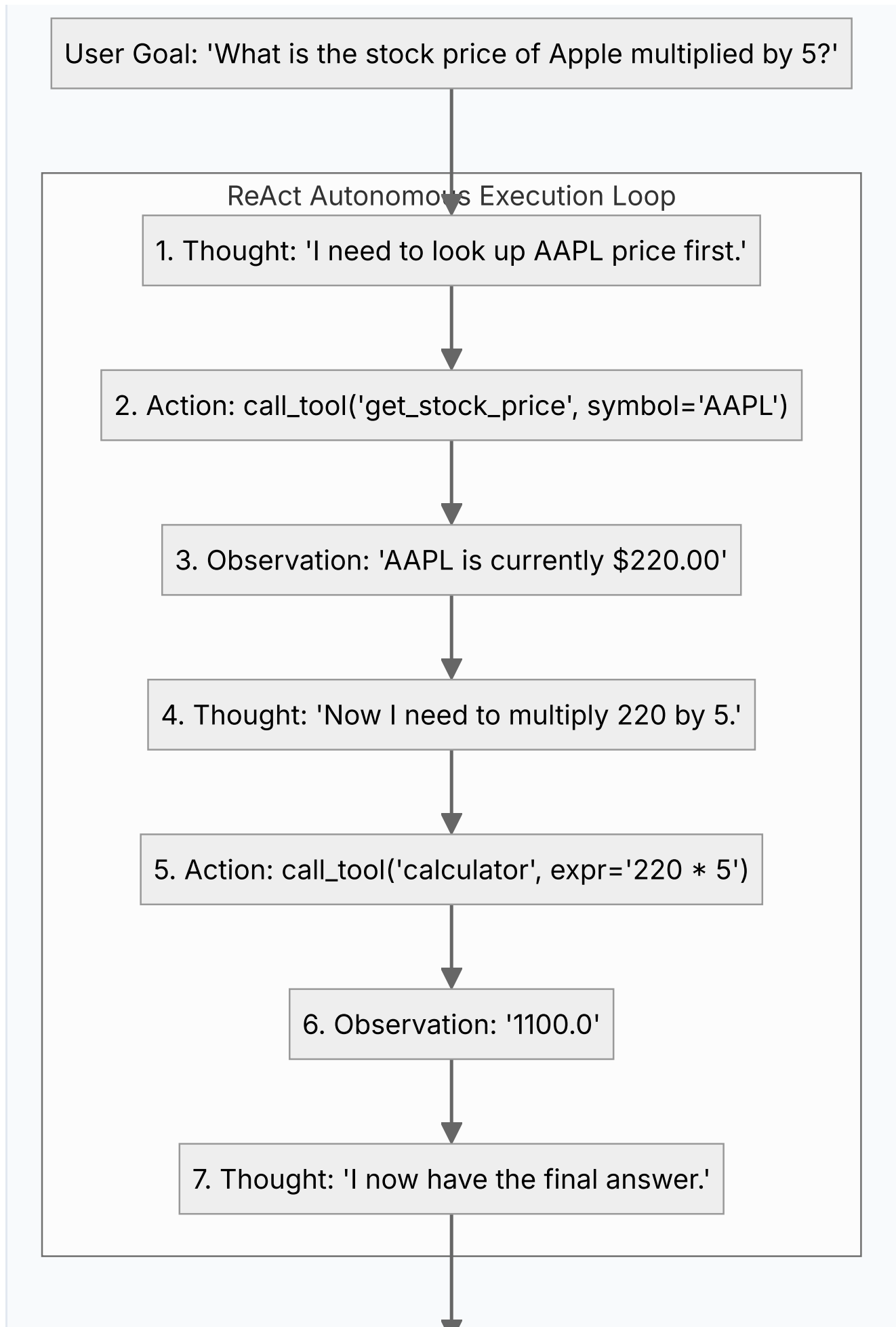
Agentic AI Chapter 4: Agent Cognitive Architectures: ReAct, Plan-and-Solve & Reflection

Core Learning Objective: Master the cognitive loops that drive autonomous AI agents. Build a working ReAct (Reasoning + Acting) loop from scratch, understand Plan-and-Solve architectures, and implement iterative self-correction with Reflexion.

1. The ReAct Pattern (Reasoning + Acting)

The **ReAct** architecture (Yao et al., 2022) combines internal reasoning traces with external tool actions in an iterative cycle:



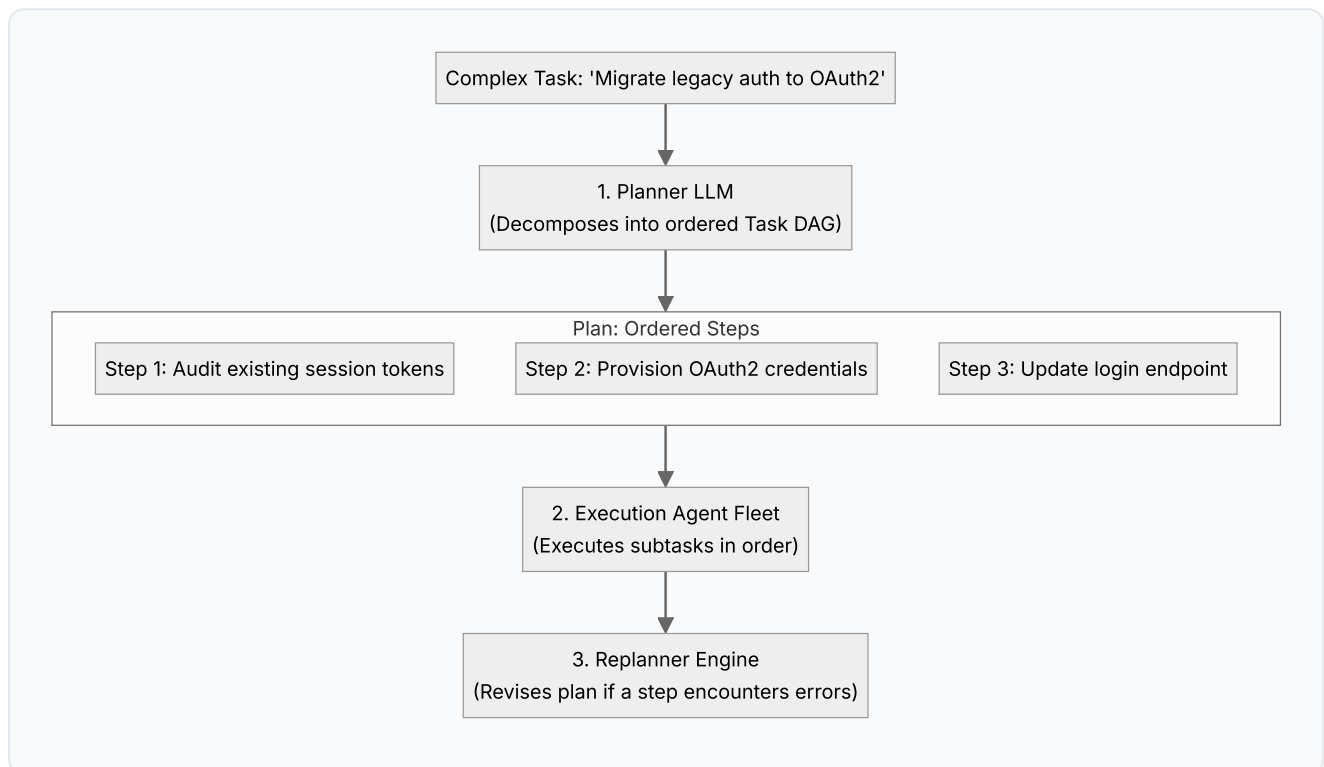


Final Response: '5 shares of Apple equal \$1,100.00'

Complete ReAct Loop Implementation in Python

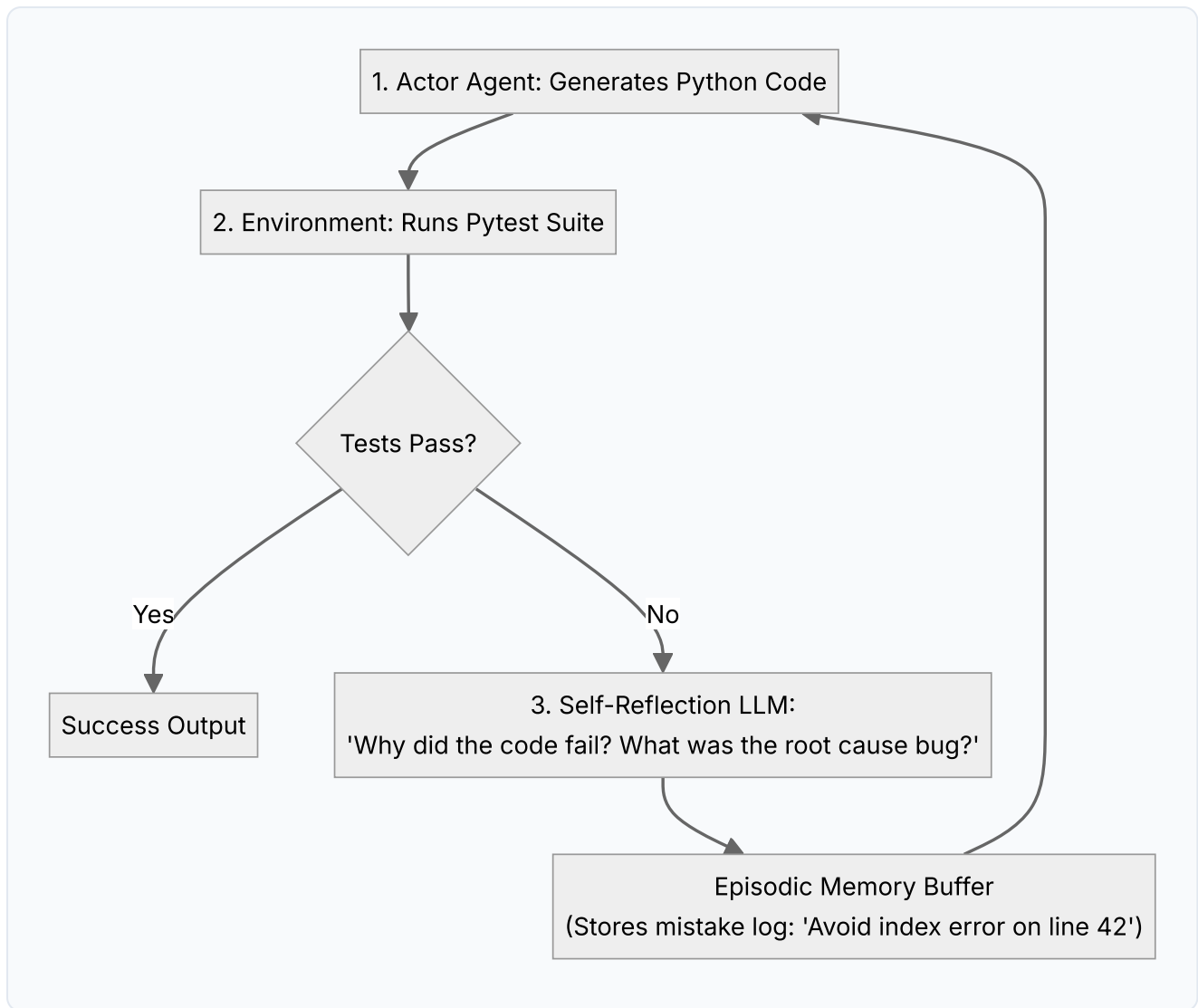
2. Plan-and-Solve Architecture

While ReAct decides each step greedily one by one (which can get lost in deep rabbit holes), **Plan-and-Solve** separates high-level planning from mechanical execution:



3. Reflexion: Iterative Self-Correction

Reflexion (Shinn et al., 2023) equips agents with an **evaluator and memory of failures**:



By reading its own previous mistakes in memory, the model refactors its code and avoids repeating bugs, increasing complex coding benchmark accuracy by > 30%.

Further Reading

- [ReAct paper](#)
- [Anthropic: building effective agents](#)
- [Lilian Weng: LLM powered autonomous agents](#)

Check Yourself

Answer in your head or on paper first, then open each answer.

- 1. ReAct in one sentence?
- 2. Plan-and-solve versus ReAct?

- ▶ 3. Why do agents need step budgets?
- ▶ 4. Why model an agent as a state machine?